Eric Spencer

Azure Migration and Integration Engineering

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HorneSci engineer with Azure identity provisioning, enterprise software delivery and IT service desk experience. Published research spans TLA+ verification and incremental sparse computing.

Technical Skills

Azure: Microsoft Azure; Microsoft Entra ID (Azure AD); access provisioning; Resource Graph export analysis
Migration: Tenant migration planning; inventory reconciliation; dependency mapping; rollback planning; acceptance evidence
Engineering: Python; Java; C++; TypeScript; SQL; Git; Docker; CI/CD; Kafka; TLA+

Experience

HorneSci | **Member of Technical Staff** Jun 2026 - Present

- Run technical discovery, scope customer requirements and deliver reusable integrations across Python, Java and C++ systems.
- Integrate HST and Kafka event pipelines; document implementation and operational handoff.

Loyola University Chicago | **ITS Service Desk Technician** Jul 2024 - May 2026

- Provisioned secure access with Azure AD and BI tools for students, faculty and staff.
- Resolved more than 50 support tickets weekly; traced incidents and coordinated escalation with infrastructure teams.

AI4FM at Loyola University Chicago | **Researcher** Aug 2025 - Present

- Build AI pipelines that invoke SANY and TLC, trace failures and validate TLA+ specifications for distributed systems.

CCC Intelligent Solutions | **Software Engineering Intern** May 2025 - Aug 2025

- Delivered Vue.js and Spring Boot changes for an enterprise insurance claims portal, including operational dashboards and QA verification within CI/CD.

EpiFinder Medical Solutions | **Software Engineering Intern** Jan 2025 - May 2025

- Automated HR data and permissions workflows with Google Apps Script, processing more than 400 email-based access requests in seconds.

Selected Project

Azure Migration Assurance | **HorneSci offline prototype** Nov 2026

- Built an offline prototype with dependency tracking, durable snapshots and selective revalidation; verified with 51 tests and 100 generated dependency graphs.

TLA and Systems Publications

TLA-Prover: Verifiable TLA+ Specification Synthesis via Preference-Optimized Low-Rank Adaptation. First author. ICSOFT 2026, pp. 627-636.
Can LLMs Write Correct TLA+ Specifications? Evaluating Natural-Language-to-TLA+ Generation. Coauthor. ICSOFT 2026, pp. 39-50.
TLA+-Bench: An Execution-Grounded Benchmark and Dataset for Natural-Language to TLA+ Specification Generation. Coauthor. Preprint, arXiv:2607.23425, 2026.
How the Dirty Set Moves: Characterizing Schedule Reuse in Incremental Sparse Matrix Multiplication. Coauthor. Systems preprint, arXiv:2610.19489, 2026.

Education

Loyola University Chicago May 2026
B.S. Computer Science, Minor in Information Systems
Dean's List; Mulcahy Scholar; Presidential Scholarship